GROUP 1:
CROP YIELD FORECAST (PROBLEM 2)

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OUR APPROACH

- Use Prithvi to analyze data to create yield forecasts
- Use Jupyter Notebooks for collaboration
- Use AWS Quick and Claude for help coding and understanding code issues

FOUNDATION MODEL DATA ANALYSIS

- Satellite Geo-Spatial Data
- Prithvi 2.0
- Python
- EVI Time Series May – September
- NASA HLS (Landsat +Sentinel 2)
- Enhanced Vegetation Index EVI
- Bushels/acre (BU/ACRE)
- Cone of Uncertainty
- t-SNE Clustering
- NASS API

OUR PROCESS

- Establish coding environment
- Go through given tutorials
- Establish model (Prithvi-EO-2. 300m)
- Test Prithvi to deepen our understanding (PrivaTest file)
- Establish initial model
- Feed model more data
- Create new Iterations and repeat
-

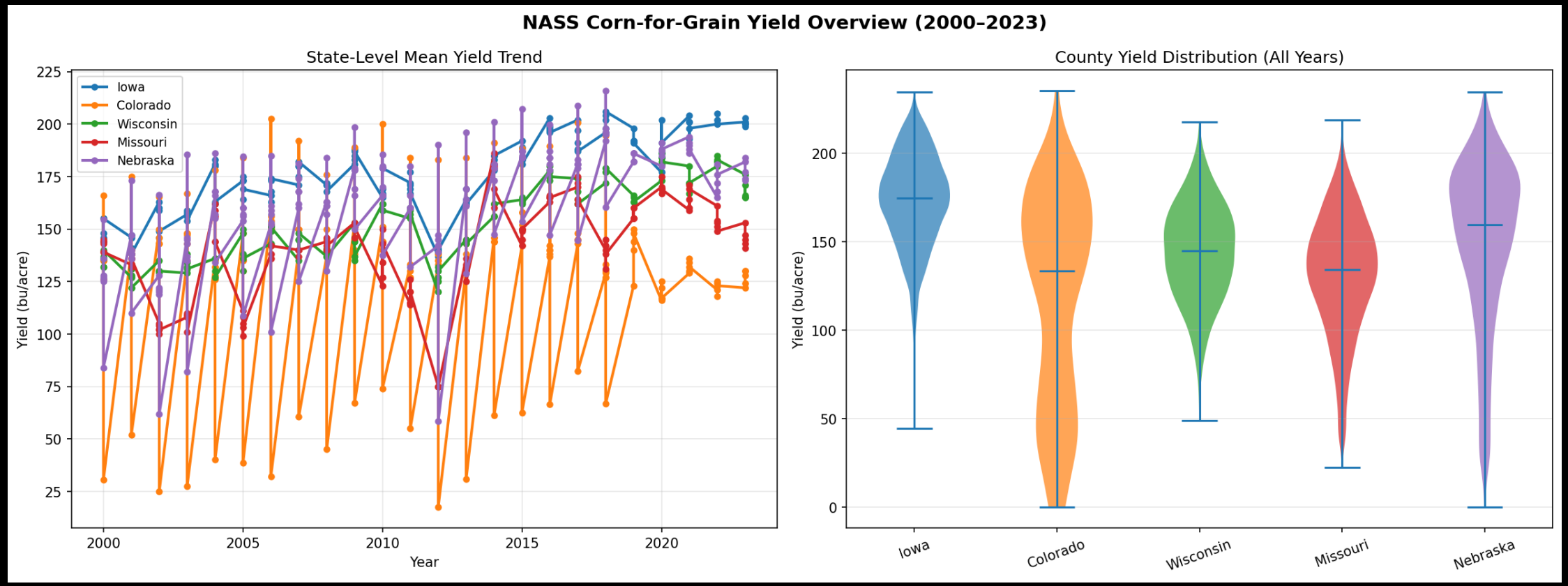
PRIVATEST FILE

- First attempt at using Prithvi 2.0
- Majority of time spent developing this version
- Use AWS environment and tools exclusively
- This notebook loads and tests the Prithvi-EO 2.0 model from IBM/NASA by feeding it multispectral satellite imagery across multiple time steps to verify it runs correctly on real geospatial data.

CORN YIELD PIPELINE FILE

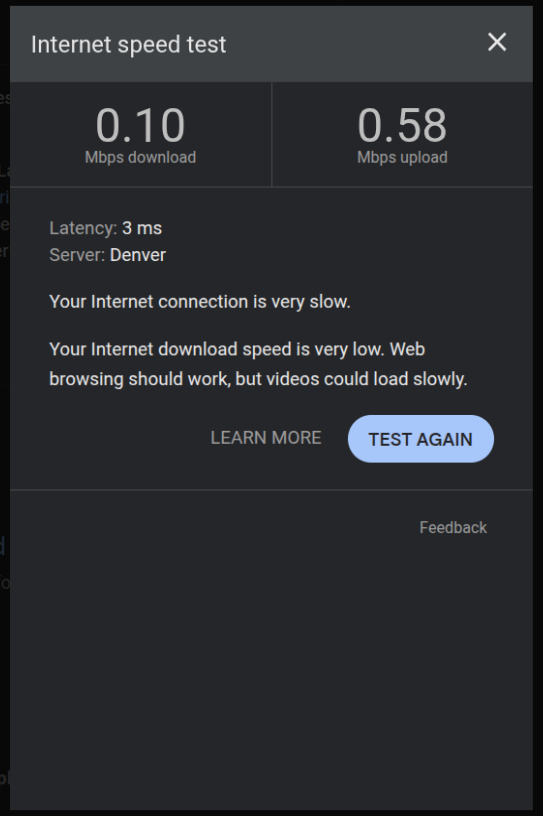
- Our pivot after being locked out of AWS
- Less than 5 hours to catchup
- This notebook uses NASA satellite imagery and a pretrained geospatial AI model to predict corn yields across five Midwest states at different points during the growing season, outputting probabilistic forecasts with uncertainty ranges and a final report.

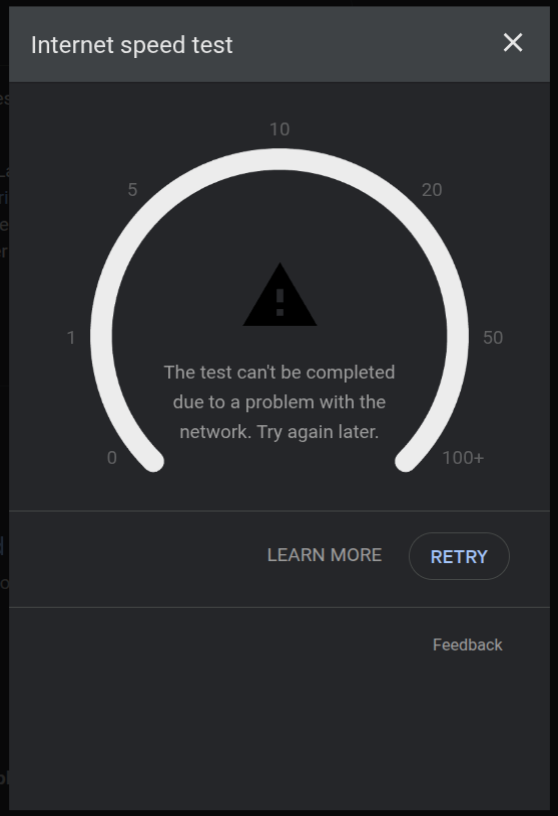
ANALYSIS AND PLOTTING OF TRAINING DATA

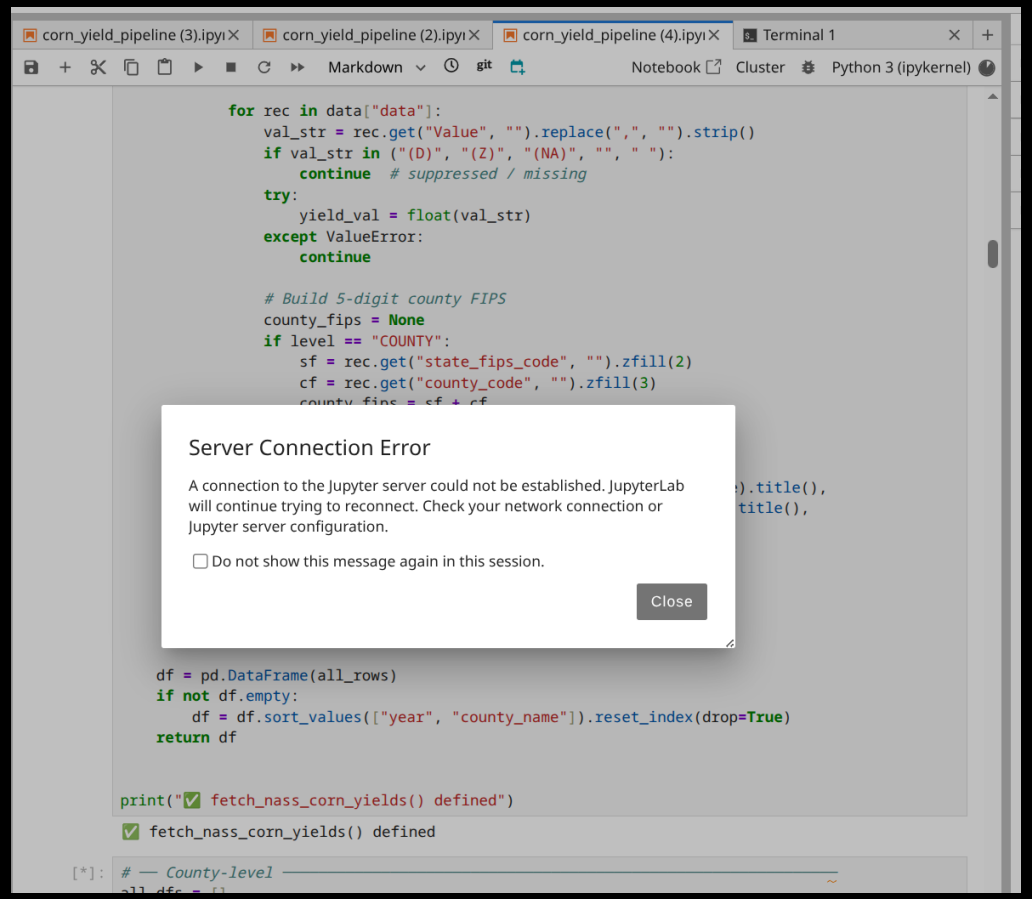


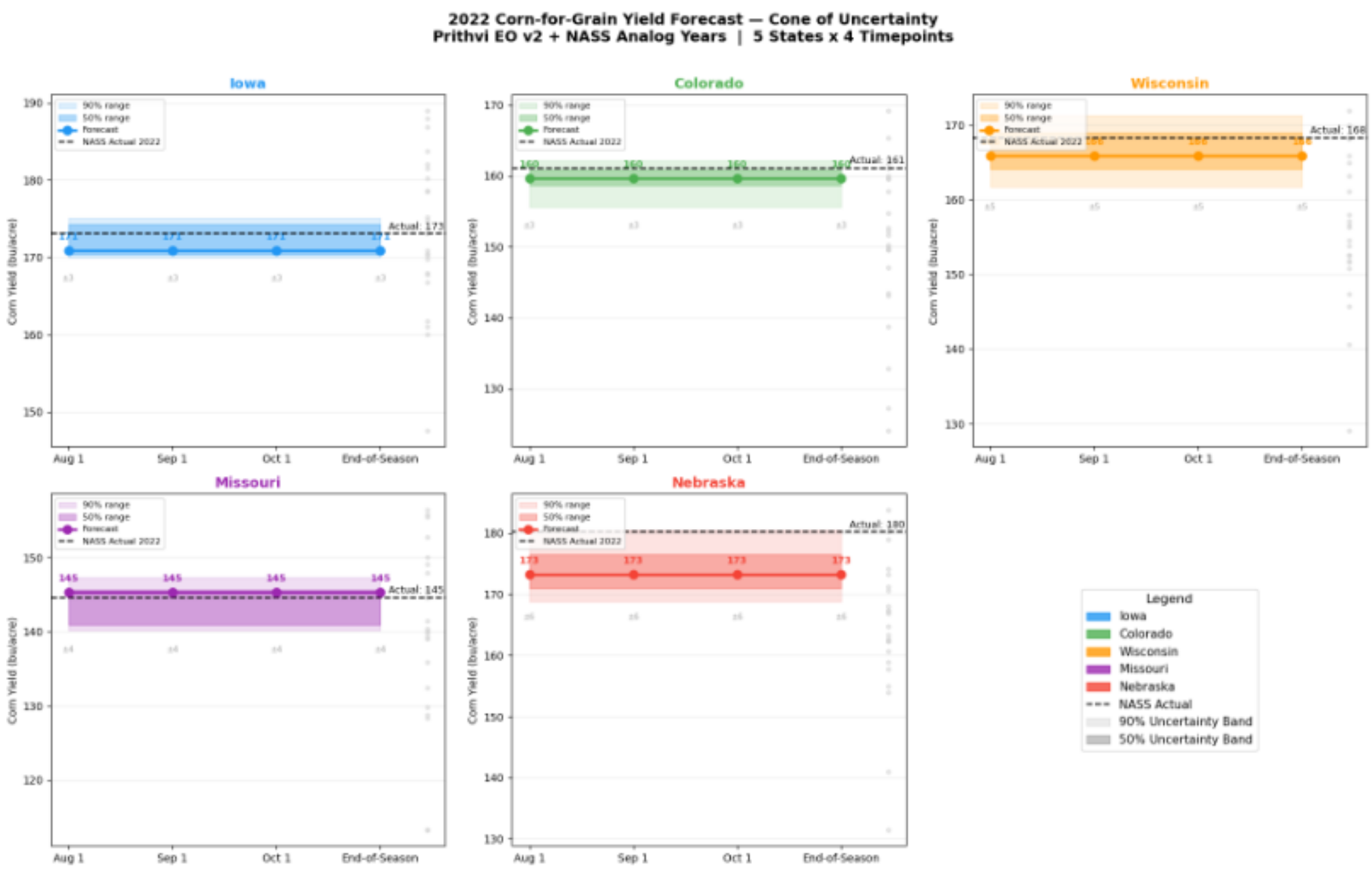
DIFFICULTIES ALONG THE WAY

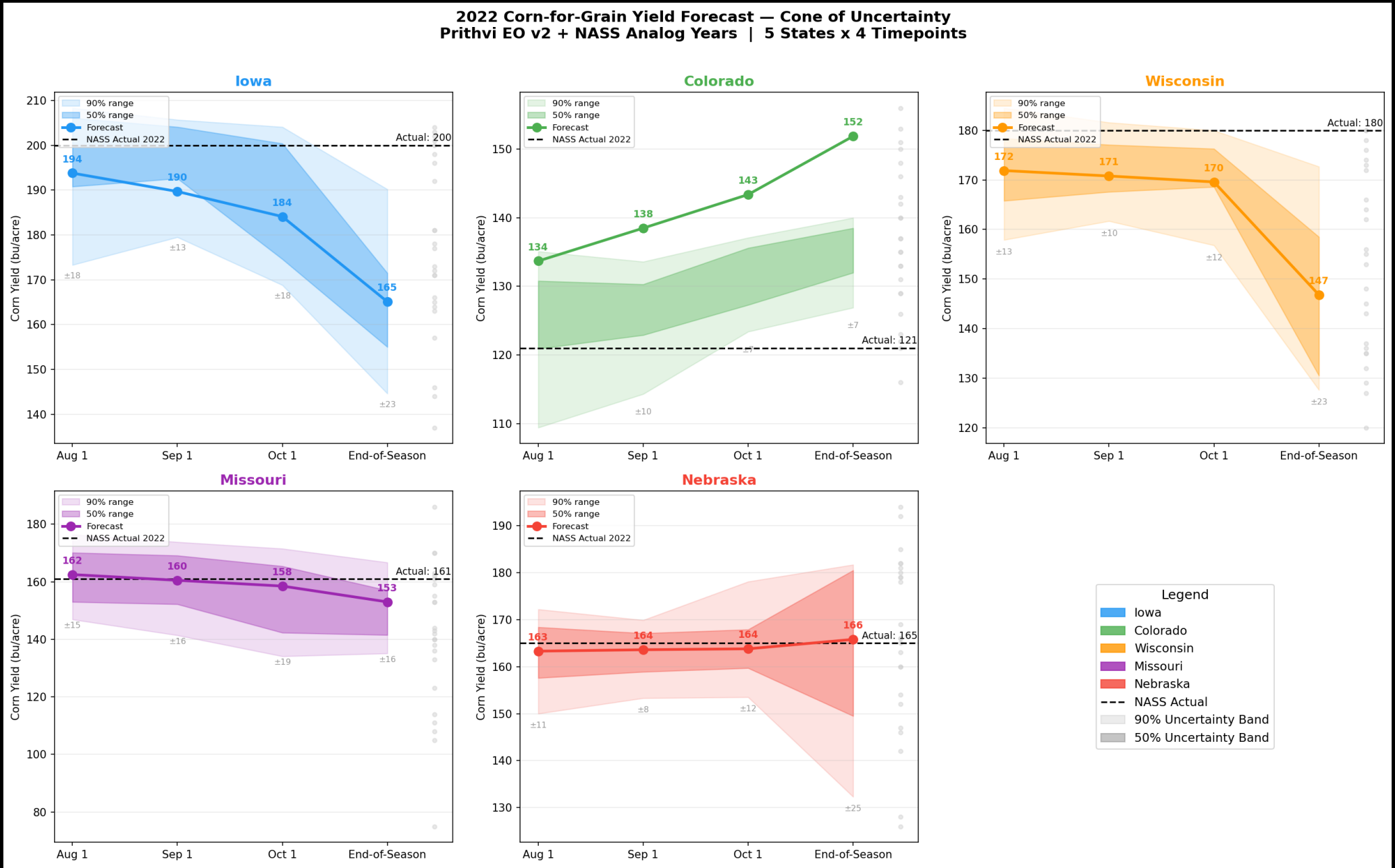
- Only 3 group members
- Network connection issues
- Kicked out of AWS services at 5AM
- Jupyter becoming inaccessible multiple times (longest 1 hour plus)
- Jupyter running out of Kernel memory
- Issues accessing RODA "buckets"


Internet speed test
0.10
Mbps download
0.58
Mbps upload
Latency: 3 ms
Server: Denver
Your Internet connection is very slow.
Your Internet download speed is very low. Web browsing should work, but videos could load slowly.
LEARN MORE
TEST AGAIN
Feedback


Internet speed test
10
5
20
1
50
0
100+
The test can't be completed due to a problem with the network. Try again later.
LEARN MORE
RETRY
Feedback


corn_yield_pipeline (3).ipyn
corn_yield_pipeline (2).ipyn
corn_yield_pipeline (4).ipyn
Terminal 1
Markdown
Notebook
Cluster
Python 3 (ipykernel)
for rec in data["data"]:
val_str = rec.get("Value", "").replace(",", "").strip()
if val_str in ("(D)", "(Z)", "(NA)", "", " "):
continue # suppressed / missing
try:
yield_val = float(val_str)
except ValueError:
continue
# Build 5-digit county FIPS
county_fips = None
if level == "COUNTY":
sf = rec.get("state_fips_code", "").zfill(2)
cf = rec.get("county_code", "").zfill(3)
Server Connection Error
A connection to the Jupyter server could not be established. JupyterLab will continue trying to reconnect. Check your network connection or Jupyter server configuration.
Do not show this message again in this session.
Close
df = pd.DataFrame(all_rows)
if not df.empty:
df = df.sort_values(["year", "county_name"]).reset_index(drop=True)
return df
print("✅ fetch_nass_corn_yields() defined")
✅ fetch_nass_corn_yields() defined
— County-level


2022 Corn-for-Grain Yield Forecast — Cone of Uncertainty
Prithvi EO v2 + NASS Analog Years | 5 States x 4 Timepoints
Iowa
Colorado
Wisconsin
Missouri
Nebraska
90% range
50% range
Forecast
NASS Actual 2022
Corn Yield (bu/acre)
Aug 1
Sep 1
Oct 1
End-of-Season
Actual: 173
Actual: 161
Actual: 168
Actual: 145
Actual: 180
Legend
Iowa
Colorado
Wisconsin
Missouri
Nebraska
NASS Actual
90% Uncertainty Band
50% Uncertainty Band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Forecast
NASS Actual 2022
Corn Yield (bu/acre)
Aug 1
Sep 1
Oct 1
End-of-Season
Actual: 200
194
190
184
165
±18
±13
±18
±23
Actual: 121
134
138
143
152
±10
±7
Actual: 180
172
171
170
147
±13
±10
±12
±23
Actual: 161
162
160
158
153
±15
±16
±19
±16
Actual: 165
163
164
164
166
±11
±8
±12
±25
Legend
Iowa
Colorado
Wisconsin
Missouri
Nebraska
NASS Actual
90% Uncertainty Band
50% Uncertainty Band

HOW TO IMPROVE

- Import data rather than scrape it to reduce wait times
- Train the model more in order to improve outputs
- Test our model on datasets other than our training data
- Possibly pivot to a "local" approach to avoid networking and environment troubles

THANK YOU !!!! :)